

HOSPITAL MANAGEMENT INFORMATION SYSTEM

Dimensional Data Modeling Project

Power BI | Kimball Dimensional Modeling

Project Report

Objective: Design and implement a business-oriented dimensional data model for hospital analytics using Power BI and Kimball dimensional modeling principles.

Business Areas Covered

Operations • Clinical • Finance • Insurance • Pharmacy • HR

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Tools: Power BI • Power Query • DAX • Excel

Dataset: Synthetic HMIS Dataset | 2020–2025

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Executive Summary

This project developed a Hospital Management Information System (HMIS) dimensional data model in Power BI using practical Kimball dimensional modeling principles.

The source dataset consists of 19 normalized tables covering hospital operations, patients, admissions, clinical diagnostics, prescriptions, billing, insurance, pharmacy inventory and staff management.

The project followed a business-process-first and grain-first approach. Source tables were profiled and classified before being transformed into a hybrid dimensional model consisting of business dimensions, transaction facts, an accumulating admission snapshot, a current-state inventory fact and process-specific flag/junk dimensions.

The final model uses a star schema as the default, with selective dimension-side relationships where they provide analytical value without unnecessary duplication or sparsity. Direct fact-to-fact relationships were avoided.

The model was validated against the source data using ground-truth KPI calculations, filter testing, relationship testing, grain validation, billing reconciliation and cross-fact isolation testing.

The resulting model provides an analytical foundation for hospital reporting across:

Operations | Clinical | Finance | Insurance | Pharmacy | HR

Key Result: the project demonstrates how a normalized operational dataset can be transformed into a validated, business-oriented dimensional model designed for reliable Power BI analysis.

1. Project Overview

1.1 Project Objective

This project focuses on designing a dimensional data model for a Hospital Management Information System (HMIS) using Power BI and Kimball dimensional modeling principles.

The objective is to transform a normalized relational hospital dataset into a clean analytical model that supports reporting and analysis across major hospital functions such as:

- Patient admissions
- Diagnostics
- Prescriptions
- Billing
- Insurance
- Pharmacy inventory
- Staff assignments

The model is designed to maintain appropriate fact-table grain, provide reusable dimensions for analysis, avoid direct fact-to-fact relationships, and support efficient filtering and reporting in Power BI.

1.2 Dataset

The project uses a synthetic but realistic HMIS dataset containing 19 source tables covering hospital operations for the period 2020–2025. The source data represents entities such as:

- Patients
- Admissions
- Departments
- Wards and beds
- Diseases
- Doctors and employees
- Diagnostic tests
- Prescriptions
- Billing
- Insurance
- Drugs and manufacturers
- Inventory
- Staff assignments

The original source tables are retained in Power Query for traceability and reference, while transformed analytical tables are created separately for the final Power BI model.

1.3 Modeling Approach

The project follows a Kimball-oriented dimensional modeling approach, with:

- Grain-first fact design
- Fact and dimension separation
- Conformed dimensions
- Business-process-based fact tables
- Aggregation before combining data at different grains
- No direct fact-to-fact relationships
- Star schema as the default design
- Selective snowflaking where flattening would create unnecessary data duplication, expansion, or sparsity

The resulting model is therefore a hybrid dimensional model: predominantly star-based, with deliberately retained dimension-side snowflake structures where they are appropriate for the HMIS domain.

1.4 Tools Used

Tool	Purpose
Power Query	Data preparation, transformation, merging and creation of analytical tables
Power BI	Dimensional model, relationships and analytical reporting
DAX	Date dimension and analytical calculations
Excel	Cross checking the data validity
Kimball methodology	Dimensional modeling framework

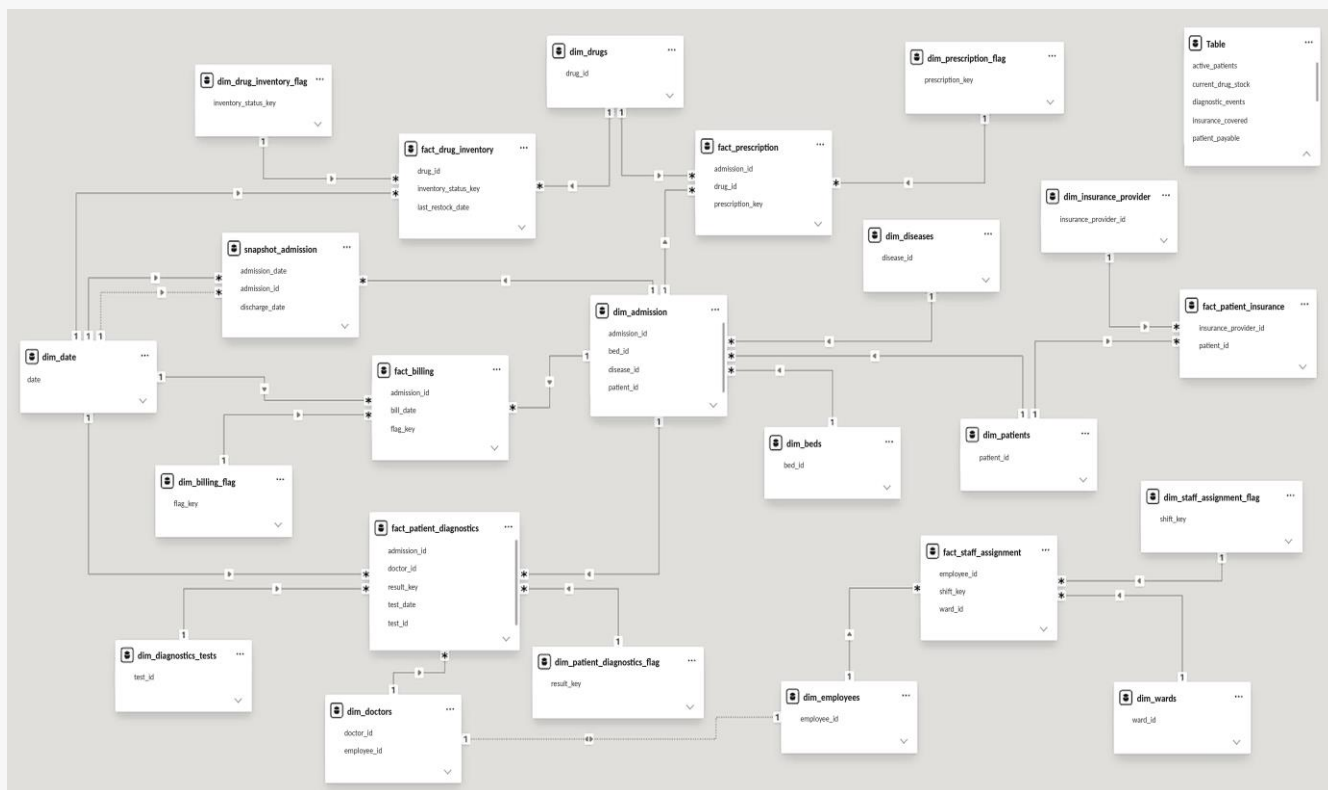


Figure 1. Final HMIS dimensional model in Power BI.

2. Source Dataset & Data Profiling

2.1 Source Dataset

The original HMIS dataset contains 19 source tables representing the major entities and operational processes of a hospital. The source tables were retained separately in Power Query and kept disabled from the final Power BI semantic model. Analytical dimensions, facts and snapshot tables were then created from these sources.

Category	Source Tables
Hospital structure	department, ward, bed
Clinical reference	disease, diagnostic_test
Pharmacy	drug, drug_manufacturer, drug_inventory
Patient & admission	patient, admission
Finance	billing, billing_detail
Clinical transactions	patient_diagnostic, prescription
Insurance	insurance_provider, patient_insurance
Staff	employee, doctor, staff_assignment

2.2 Source Table Sizes

Initial profiling was performed to understand the scale of each source table.

Source Table	Rows
department	11
ward	27
bed	415
disease	20
diagnostic_test	9
drug_manufacturer	300
insurance_provider	50
patient	30,000
admission	45,000
billing	45,000
billing_detail	112,402
patient_diagnostic	63,269
drug	250
drug_inventory	250

Source Table	Rows
prescription	73,109
patient_insurance	21,617
employee	500
doctor	98
staff_assignment	207

2.3 Data Profiling

Initial Source Table Classification and Grain

Before designing the dimensional model, all 19 source tables were reviewed individually to understand their business purpose, level of detail and potential role in the analytical model. The grain of each source table was identified before deciding whether it should become a dimension, fact, snapshot or be transformed into another analytical structure.

#	Source Table	Initial Classification	Source Grain
1	department	Dimension	1 row per department
2	ward	Dimension	1 row per ward
3	bed	Dimension	1 row per bed
4	disease	Dimension	1 row per disease
5	diagnostic_test	Dimension	1 row per diagnostic test
6	drug_manufacturer	Dimension / Reference	1 row per manufacturer
7	insurance_provider	Dimension	1 row per insurance provider
8	patient	Dimension	1 row per patient
9	admission	Fact Candidate / Encounter	1 row per admission
10	billing	Fact	1 row per bill
11	billing_detail	Fact Candidate	1 row per billing detail
12	patient_diagnostic	Fact	1 row per diagnostic event
13	drug	Dimension	1 row per drug
14	drug_inventory	Snapshot Fact Candidate	1 row per drug inventory record
15	prescription	Fact	1 row per prescription
16	patient_insurance	Fact	1 row per patient insurance policy
17	employee	Dimension	1 row per employee
18	doctor	Dimension	1 row per doctor
19	staff_assignment	Fact	1 row per staff assignment

Recognizing these grains became important later when designing the fact tables and the admission-level snapshot.

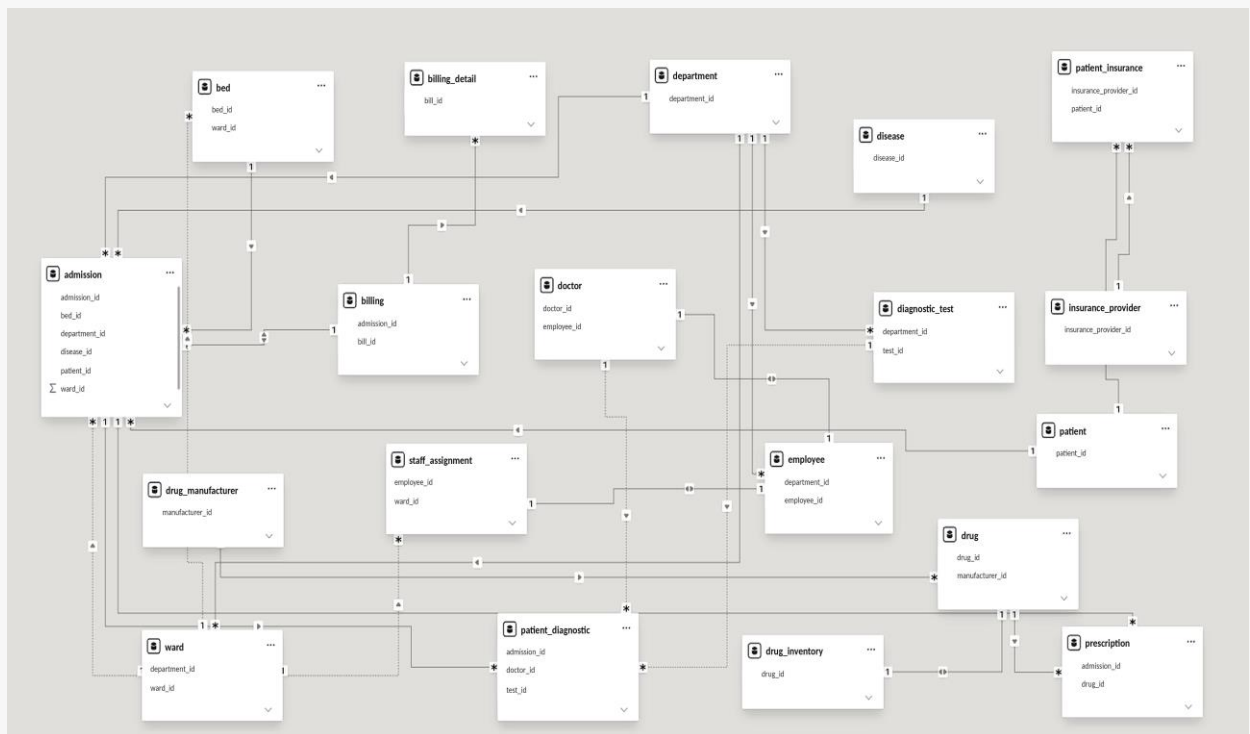


Figure 2. Original Source Model.

Key takeaway: The original default Power BI Model is messy and often inaccurate, it must be designed properly.

2.4 Power Query Organization

The 19 source tables resulted in more queries in Power Query because some source tables were referenced to create transformed analytical tables. The Power Query structure was organized into:

SOURCE → DIMENSIONS → FACTS → SNAPSHOT-FACTS → OTHER QUERIES

The original source queries were retained for traceability but disabled from the final model. Only the cleaned analytical tables were enabled for the Power BI semantic model.

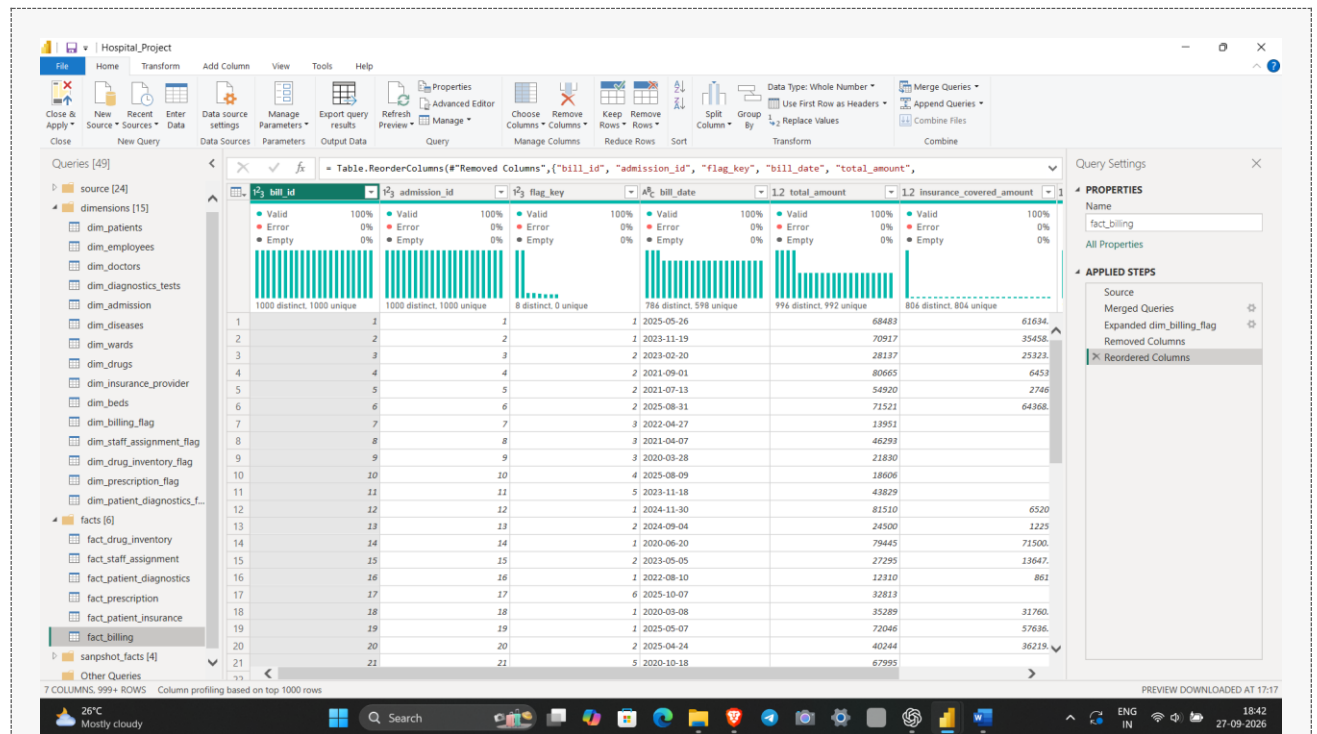


Figure 3. Power Query organization of source and analytical queries.

Key takeaway: the profiling stage established the structure, grain and relationships of the 19 source tables before dimensional modeling began.

3. Business Requirements

The dimensional model was designed around the major analytical needs of a hospital rather than around the structure of the source CSV files. The requirements were considered from the perspective of the following hospital functions:

Business Function	Key Analytical Requirements
Hospital Operations	Admissions, admission types/status, departments, wards, beds and length of stay
Clinical Management	Diseases, diagnostic tests, test outcomes and doctor activity
Finance	Billing amounts, insurance-covered amounts, patient payable amounts, payment status and payment mode
Insurance	Patients covered, insurance providers, policy information and coverage percentage
Pharmacy / Inventory	Drugs, manufacturers, current stock, reorder levels and inventory status
HR / Staff Management	Employees, roles, doctors, ward assignments and shifts
Hospital Management	Cross-functional analysis using common dimensions such as date, patient, admission, drug and organizational structures

3.1 Major Business Processes

Based on these requirements, the following processes were identified for dimensional modeling:

1. Patient Admissions
2. Billing
3. Diagnostic Testing
4. Prescriptions
5. Patient Insurance
6. Drug Inventory
7. Staff Assignment
8. Patient Journey / Admission Lifecycle

Each process was subsequently analyzed separately to determine its appropriate fact grain.

3.2 Important Data Limitations Identified

The requirements analysis also exposed several limitations in the available dataset.

Area	Limitation
Bed utilization	No historical bed-occupancy history
Inventory	Only current stock information; no stock-movement history
Staff	No assignment date/history for historical staffing analysis
Insurance	No reliable admission/bill-level policy linkage

Area	Limitation
Billing detail	billing_detail amounts did not directly reconcile with billing.total_amount
Prescriptions	No prescription date in the source
Admission lifecycle	Some journey milestones therefore have to be derived only where source data supports them

These limitations were treated as data/modeling constraints rather than assumptions to be filled artificially.

3.3 Requirement-to-Model Principle

Business requirements determine the analytical model; the source-table structure does not automatically determine the final Power BI model.

This meant that source tables were evaluated according to their business process, grain, relationships and analytical usefulness before being converted into dimensions or facts.

Section takeaway: the requirements analysis established the business processes to be modeled and identified important data limitations before the dimensional design was finalized.

4. Dimensional Modeling Approach

The HMIS model was designed using Kimball dimensional modeling principles, with the goal of creating a model that is easy to understand, analyze, and extend.

4.1 Core Modeling Principles

Principle	How It Was Applied
Grain first	The grain of every fact, snapshot, and dimension was defined before building the table.
Business-process driven	Fact tables were designed around hospital processes such as admissions, billing, diagnostics, prescriptions, and staff assignments.
Facts vs Dimensions	Transactional measurements/events were separated from descriptive analytical context.
Conformed dimensions	Common dimensions such as Patient, Date, Admission, and other shared entities provide consistent filtering across processes.
No fact-to-fact relationships	Facts are analyzed independently through shared dimensions rather than directly joining facts.
Aggregate before merge	Data at a lower grain was aggregated before being incorporated into a higher-grain table to prevent duplication and fan-out and numbers were protected.
Star schema as the default	Dimensions were flattened where this improved usability without creating unnecessary duplication.
Selective snowflake	Dimension-to-dimension relationships were retained only where flattening would cause excessive expansion, NULLs, repetition, or grain mismatch.
Source vs analytical layer	Original source queries were retained for traceability, while separate transformed tables form the analytical model.
Source IDs retained where appropriate	Since this is a stable single-source dataset, original business keys were retained for most dimensions rather than introducing unnecessary surrogate keys.

4.2 Overall Modeling Structure

The final model is a hybrid dimensional model: star schema wherever practical + selective dimension-side snowflake where justified.

For example:

- fact_billing → dimensions
- fact_patient_diagnostics → dimensions
- fact_prescription → dimensions
- fact_staff_assignment → dimensions
- fact_drug_inventory → dimensions
- snapshot_admission → dimensions
- dim_admission provides shared admission-level analytical context

- Selected dimension-to-dimension relationships are retained where they better represent the business structure

The objective was not to force every table into a pure star schema, but to balance analytical simplicity with the actual structure and grain of the hospital data.

4.3 Key Modeling Rule

Define the grain → identify dimensions → identify facts → establish relationships → validate the model.

This approach helped avoid common problems such as duplicated measures, fan-out, ambiguous relationships, and inappropriate fact-to-fact joins.

Section takeaway: the model follows Kimball principles while using a controlled hybrid approach rather than forcing the HMIS data into a purely star-shaped structure.

5. Dimension Design Decisions

Dimensions were designed based on business meaning, grain, analytical usage, and relationship structure, rather than simply converting every source table into a dimension.

5.1 Final Dimension Design

Dimension	Grain	Design Decision
dim_patients	1 row per patient	Retained as the central patient dimension.
dim_admission	1 row per admission	Created as a shared encounter dimension containing admission-level descriptive information.
dim_beds	1 row per bed	Bed, ward, and department information flattened at bed grain.
dim_wards	1 row per ward	Ward and department information flattened at ward grain. Kept separately because staff assignments operate at ward grain.
dim_diseases	1 row per disease	Disease reference information retained as a dimension.
dim_diagnostic_tests	1 row per diagnostic test	Test and relevant department information flattened together.
dim_drugs	1 row per drug	Drug and manufacturer information flattened at drug grain.
dim_insurance_provider	1 row per provider	Provider information retained as a dimension.
dim_employees	1 row per employee	Employee and department information flattened together.
dim_doctors	1 row per doctor	Kept separate because doctors represent a specialized subset of employees.
dim_date	1 row per date	Conformed date dimension used across relevant business processes.

5.2 Flag / Junk Dimensions

In addition to the main business dimensions, several small flag/junk dimensions were created to handle low-cardinality attributes associated with individual fact processes. These tables follow the same dimensional principle of separating descriptive filtering attributes from the transactional fact while avoiding unnecessary descriptive columns directly inside the fact table.

The final model contains the following flag dimensions:

Flag Dimension	Connected Fact	Purpose
dim_billing_flag	fact_billing	Stores billing-related flag attributes
dim_drug_inventory_flag	fact_drug_inventory	Stores inventory-status-related flag attributes
dim_patient_diagnostics_flag	fact_patient_diagnostics	Stores diagnostic-result-related flag attributes
dim_prescription_flag	fact_prescription	Stores prescription-related flag attributes

Flag Dimension	Connected Fact	Purpose
dim_staff_assignment_flag	fact_staff_assignment	Stores staff-assignment/shift-related flag attributes

Each fact contains a corresponding key such as flag_key, result_key, prescription_key, shift_key, or the relevant inventory key, which connects the fact to its small flag dimension.

Why Were Flag Dimensions Used?

These dimensions were useful because the associated attributes are:

- Low-cardinality
- Used mainly for filtering/grouping
- Specific to a particular business process
- Not sufficiently important to justify creating large standalone business dimensions

For example, diagnostic result categories can be handled through dim_patient_diagnostics_flag, while staff shift categories can be handled through dim_staff_assignment_flag. This keeps the fact tables focused primarily on their business-process grain and measures, while providing a clean dimensional structure for filtering.

Maintenance Consideration

Flag/junk dimensions require maintenance when new combinations appear in future source data. If a new combination of flag values appears that does not exist in the corresponding flag table, the dimension needs to be updated so that the new records can be correctly classified and filtered.

Therefore, flag dimensions are part of the model's ETL/data-refresh responsibility rather than static tables that can be ignored after initial creation.

5.3 Important Design Decisions

1. Flattening where appropriate

Related descriptive tables were combined when they represented the same analytical entity. Examples:

- Drug + Drug Manufacturer → dim_drugs
- Ward + Department → dim_wards
- Bed + Ward + Department → dim_beds
- Employee + Department → dim_employees
- Diagnostic Test + Department → dim_diagnostic_tests

This reduces unnecessary joins during analysis.

2. Keeping dimensions separate where grain or business meaning differs

dim_wards and dim_beds were intentionally kept separate: dim_wards is one row per ward, dim_beds is one row per bed. Flattening everything into one structure would unnecessarily repeat ward-level information for every bed and would not serve the different analytical grains cleanly.

3. Doctor dimension kept separate from Employee

Although every doctor is an employee, dim_doctors was retained separately because doctors are a specialized subset with attributes such as specialization, qualification and experience. This avoids unnecessarily widening the employee dimension with doctor-specific attributes and NULL values for non-doctors.

4. Admission as a real analytical dimension

dim_admission represents the hospital encounter/case at one row per admission. It provides a common analytical context for facts such as billing, diagnostics, prescriptions, and the admission journey without creating direct fact-to-fact relationships.

5. Creating junk/flag dimensions

For low-cardinality attributes associated with specific business processes, instead of treating these attributes as unrelated standalone dimensions or adding excessive descriptive columns to fact tables, dedicated flag tables were created and connected to their corresponding facts (see Section 5.2).

6. Selective dimension-side snowflake

Some dimension-to-dimension relationships were intentionally retained where flattening would create unnecessary expansion, repetition, or grain problems. Therefore, the model uses controlled snowflaking rather than eliminating all dimension-to-dimension relationships.

5.4 Key Principle

A source table does not automatically become a dimension. The final dimension structure is determined by analytical grain and business usage.

Section takeaway: dimensions were flattened when this improved analytical usability, while separate dimensions and selective snowflake relationships were retained where they better represented the business grain and avoided unnecessary model complexity.

6. Fact Table Design

Fact tables were designed around business processes, with the grain defined before selecting measures and relationships. Each fact represents a specific analytical event or state and connects to relevant dimensions.

6.1 Final Fact Tables

Fact Table	Grain	Type	Main Purpose
snapshot_admission	1 row per admission	Accumulating snapshot	Tracks the admission lifecycle using milestone dates and admission-level counts.
fact_billing	1 row per bill	Transaction fact	Analyzes billing amounts, insurance coverage, patient payable amounts and payment information.
fact_patient_diagnostics	1 row per diagnostic event	Transaction / factless-style fact	Analyzes diagnostic activity, tests, doctors and result status.
fact_prescription	1 row per prescription	Transaction fact	Analyzes prescribed drugs, dosage, frequency and duration.
fact_patient_insurance	1 row per insurance policy record	Transaction/fact table	Analyzes patient insurance coverage and policy information.
fact_drug_inventory	1 row per drug inventory record	Current-state snapshot	Represents the available inventory state, including stock and reorder information.
fact_staff_assignment	1 row per assignment	Factless fact	Analyzes employee-to-ward assignments and shifts.

6.2 Admission Journey as an Accumulating Snapshot

snapshot_admission was designed as an accumulating snapshot because an admission progresses through identifiable stages. It contains:

- Admission date
- First/last diagnostic dates where available
- Discharge date
- Other supported milestone dates
- Number of diagnostic tests
- Number of prescriptions

Child-level data was aggregated to admission grain before being merged into the snapshot. This prevents multiple diagnostic or prescription records from multiplying the admission row.

Important: billing measures were not placed in the admission snapshot. Billing remains independently represented in fact_billing.

6.3 Billing Fact

fact_billing remains at bill grain. Key measures include:

- total_amount
- insurance_covered_amount
- patient_payable_amount

Payment status and payment mode provide additional analytical context. billing_detail was not incorporated into this fact because its grain is different and its amounts did not directly reconcile with the billing totals.

6.4 Other Transaction Facts

The remaining business processes retain their natural transactional grain:

- One diagnostic event → fact_patient_diagnostics
- One prescription → fact_prescription
- One insurance policy record → fact_patient_insurance
- One staff assignment → fact_staff_assignment

This preserves the original business events without artificially combining different grains.

6.5 Inventory Snapshot

fact_drug_inventory represents current inventory state, not historical inventory movement. The source contains current stock, reorder level, inventory status and last restock date, but does not contain inventory movement history. Therefore, historical stock trends were not inferred.

6.6 Key Fact Modeling Rules

1. One fact = one clearly defined grain.
2. Measures belong to the process where they originate.
3. Different grains are aggregated before being combined.
4. Facts are not directly connected to other facts.
5. Shared dimensions provide the analytical paths between business processes.
6. Source data limitations are respected rather than filled with assumptions.

Section takeaway: the fact layer separates hospital business processes by grain while using transaction facts, an accumulating snapshot, a current-state snapshot, and factless-style facts where appropriate.

7. Key Modeling Decisions & Challenges

Several modeling decisions required balancing Kimball principles, source structure, analytical requirements, and data limitations.

7.1 Major Decisions

Challenge	Decision	Reason
Different grains across hospital processes	Separate fact tables	Prevents duplication and incorrect aggregation.
Admission-related facts	Created dim_admission	Provides shared admission-level analytical context without fact-to-fact relationships.
Admission lifecycle	Created snapshot_admission	Accumulating snapshot captures the progression of an admission.
Bed, ward and department hierarchy	Selective flattening	dim_beds and dim_wards serve different grains and analytical requirements.
Doctor vs employee	Separate dimensions	Doctors are a specialized employee subset with additional attributes.
Drug + manufacturer	Flattened into dim_drugs	Manufacturer describes the drug and can be analyzed at drug grain.
Diagnostic test + department	Flattened into dim_diagnostic_tests	Keeps relevant test information together at test grain.
Billing vs billing detail	Kept separate; billing detail excluded from semantic model	Different grain and lack of direct reconciliation with billing totals.
Source IDs vs surrogate keys	Source IDs retained for most dimensions	Single stable source and no current SCD/multi-source identity requirement.
Historical inventory	Not modeled as historical	Source contains current stock only, not movement history.
Historical staff assignment	Not inferred	Assignment table has no assignment date/history.

7.2 Introduction of Flag/Junk Dimensions

Another modeling decision was the use of small flag/junk dimensions for low-cardinality attributes associated with specific business processes. Instead of treating these attributes as unrelated standalone dimensions or adding excessive descriptive columns to fact tables, dedicated flag tables were created and connected to their corresponding facts. This provided a consistent dimensional filtering mechanism while keeping the main business dimensions focused on entities such as patients, admissions, drugs, doctors and employees.

The final model therefore contains both:

- Core business dimensions — patients, admissions, drugs, doctors, etc.
- Small process-specific flag dimensions — billing, diagnostics, inventory, prescriptions and staff assignments.

7.3 Avoiding Fan-Out

One of the key challenges was combining data that existed at different grains. For example, one admission relates to multiple diagnostics and multiple prescriptions. Directly merging both child tables into admission would produce combinations such as:

3 diagnostics × 4 prescriptions = 12 rows — which would incorrectly multiply the admission-level information.

The solution was to aggregate each child process to admission grain first, then merge the aggregated results:

Diagnostics → GROUP BY admission_id → test_count → Prescriptions → GROUP BY admission_id → prescription_count

Admission → + test_count → + prescription_count → snapshot_admission

7.4 Preventing Fact-to-Fact Relationships

Another important challenge was deciding how different processes should interact. Instead of chaining Billing → Admissions → Diagnostics directly, the model uses shared analytical dimensions so that each fact remains independent. This allows analysis across processes without physically joining fact tables and risking duplicated measures.

7.5 Controlled Snowflake Design

The project initially aimed for a clean star schema. However, some dimension-side relationships were retained where flattening would create unnecessary repetition, expansion, NULLs, or grain mismatch.

The final model is intentionally a hybrid dimensional model — star schema by default, selective snowflake where justified.

Section takeaway: the major modeling challenges were resolved by applying grain-first design, aggregation before merging, independent facts, and selective dimension-side snowflaking rather than forcing every source table into a simple star.

8. Data Quality & Business Rule Decisions

Data quality checks were performed before finalizing the analytical model. The objective was to distinguish actual data issues from legitimate characteristics of the source dataset.

8.1 Key Data Quality Checks

Check	Result / Decision
Primary-key uniqueness	Checked across source tables.
Foreign-key validity	Documented relationships were validated against their parent tables.
Duplicate records	Checked during profiling before determining table grain.
Null values	Reviewed to distinguish legitimate NULLs from potential data-quality issues.
Table grain	Verified before assigning tables as facts or dimensions.
Cross-table relationships	Validated before creating model relationships.
Measure reconciliation	Billing and billing-detail amounts were compared.

8.2 Important Business Rules / Findings

Billing vs Billing Detail

billing_detail.amount does not directly reconcile with billing.total_amount. Therefore:

- billing.total_amount is used as the official bill-level amount.
- Billing-detail amounts were not treated as revenue.
- billing_detail was excluded from the semantic model for the current requirements.

It can be reintroduced later if charge-level analysis becomes a business requirement.

Insurance

patient_insurance contains policy start and end dates, but the source does not provide a reliable admission-level or bill-level policy linkage. Therefore, the model does not invent a relationship between individual bills/admissions and insurance policies.

Inventory

drug_inventory contains current stock information and last restock date, but no inventory movement history. Therefore, it is modeled as a current-state snapshot, not as a historical inventory trend fact.

Staff Assignment

staff_assignment contains employee, ward and shift information but no assignment date. Therefore, historical staffing trends were not inferred.

Prescription Dates

The prescription source does not contain a prescription date. Therefore, no artificial prescription date was created.

8.3 Data Quality Principle

Where the source does not provide sufficient evidence for a business rule, the model does not invent one.

This was particularly important for insurance linkage, historical inventory, staffing history and billing-detail interpretation.

Section takeaway: data-quality validation was used not only to find errors, but also to identify where the source data could — and could not — support analytical conclusions.

9. Date Dimension

A dedicated dim_date was created as a conformed dimension to provide consistent time-based analysis across the relevant business processes.

9.1 Design

The date dimension contains only the fields required for practical analysis:

Column	Purpose
date	Actual calendar date
day	Day number
day_name	Day of the week
month	Month number
month_name	Month name
year	Calendar year
fiscal_year	April–March financial year
fiscal_quarter	Financial quarter

9.2 Fiscal Calendar

Since the project uses an Indian financial-year structure:

- Q1: April–June
- Q2: July–September
- Q3: October–December
- Q4: January–March

For example: April 2024 → FY2024-25, Q1; January 2025 → FY2024-25, Q4.

9.3 Date Range

The date dimension was intentionally limited to the operational date range of the hospital data using CALENDAR() function by inspecting earliest and latest business related dates rather than using CALENDARAUTO() blindly. This avoids extending the calendar into unrelated historical years because of attributes such as patient dates of birth.

9.4 Date Relationships

The date dimension is connected only to business processes where date-based analysis is meaningful and supported by the source. Where a fact contains multiple date roles, such as admission and discharge dates, the model must distinguish between the relevant date relationships rather than creating multiple simultaneously active paths that could cause ambiguity.

fact_patient_insurance was intentionally not connected to dim_date in the final model.

10. Current Data Model

The final HMIS model follows a hybrid dimensional modeling approach, combining a star schema with a small number of deliberately retained dimension-side relationships where flattening would create unnecessary duplication, sparsity, or table expansion.

10.1 Model Structure

The model contains:

- Conformed dimensions shared across multiple business processes, such as dim_patients, dim_admission, dim_date, dim_drugs, dim_employees, and dim_wards.
- Business-process fact tables representing admissions-related activities, billing, diagnostics, prescriptions, insurance, inventory, and staff assignments.
- snapshot_admission as an accumulating snapshot representing the admission lifecycle.
- Flag dimensions for compact categorical/status attributes.
- Dimension-to-dimension relationships where they were deliberately retained for analytical usability.
- No direct fact-to-fact relationships.

10.2 Main Analytical Tables

Table	Grain	Purpose
dim_patients	1 row per patient	Patient analysis
dim_admission	1 row per admission	Shared admission/encounter context
dim_beds	1 row per bed	Bed → ward → department context
dim_wards	1 row per ward	Ward-level analysis
dim_diseases	1 row per disease	Disease analysis
dim_doctors	1 row per doctor	Doctor analysis
dim_employees	1 row per employee	Employee analysis
dim_diagnostic_tests	1 row per diagnostic test	Diagnostic test analysis
dim_drugs	1 row per drug	Drug/manufacturer analysis
dim_insurance_provider	1 row per provider	Insurance provider analysis
dim_date	1 row per date	Time analysis
fact_billing	1 row per bill	Financial analysis
fact_patient_diagnostics	1 row per diagnostic event	Diagnostic activity
fact_prescription	1 row per prescription	Prescription activity
fact_patient_insurance	1 row per policy	Insurance coverage analysis
fact_drug_inventory	1 row per current inventory record	Current stock analysis

Table	Grain	Purpose
fact_staff_assignment	1 row per assignment	Staff/ward assignment analysis
snapshot_admission	1 row per admission	Admission lifecycle analysis

10.3 Relationship Structure and Cardinality

The final model uses one-to-many relationships from dimensions to facts wherever applicable, with single-direction filter propagation as the default. The model also contains selected dimension-to-dimension relationships and inactive relationships where required by the analytical design.

Core relationships:

From	To	Cardinality	Status	Purpose
dim_admission	dim_beds	* : 1	Active	Connect admission to bed context
dim_admission	dim_diseases	* : 1	Active	Connect admission to disease
dim_admission	dim_patients	* : 1	Active	Connect admissions to patients
dim_admission	snapshot_admission	1 : *	Active	Connect admission context to admission snapshot
dim_doctors	dim_employees	1 : 1	Inactive	Represent doctor as specialized employee
fact_billing	dim_admission	* : 1	Active	Billing by admission
fact_billing	dim_date	* : 1	Active	Billing by bill date
fact_drug_inventory	dim_drugs	* : 1	Active	Inventory by drug
fact_drug_inventory	dim_date	* : 1	Active	Inventory by last restock date
fact_patient_diagnostics	dim_admission	* : 1	Active	Diagnostics by admission
fact_patient_diagnostics	dim_doctors	* : 1	Active	Diagnostics by doctor
fact_patient_diagnostics	dim_diagnostic_tests	* : 1	Active	Diagnostics by test
fact_patient_diagnostics	dim_date	* : 1	Active	Diagnostics by test date
fact_patient_insurance	dim_patients	* : 1	Active	Insurance policies by patient
fact_patient_insurance	dim_insurance_provider	* : 1	Active	Insurance by provider
fact_prescription	dim_admission	* : 1	Active	Prescriptions by admission
fact_prescription	dim_drugs	* : 1	Active	Prescriptions by drug

From	To	Cardinality	Status	Purpose
fact_staff_assignment	dim_employees	* : 1	Active	Staff assignment by employee
fact_staff_assignment	dim_wards	* : 1	Active	Staff assignment by ward

10.4 Flag/Junk Dimension Relationships

The flag dimensions are also part of the final semantic model and therefore should be explicitly documented.

From Fact	Flag Dimension	Key	Cardinality	Status
fact_billing	dim_billing_flag	flag_key	* : 1	Active
fact_drug_inventory	dim_drug_inventory_flag	Inventory flag key	* : 1	Active
fact_patient_diagnostics	dim_patient_diagnostics_flag	result_key	* : 1	Active
fact_prescription	dim_prescription_flag	prescription_key	* : 1	Active
fact_staff_assignment	dim_staff_assignment_flag	shift_key	* : 1	Active

The relationship pattern is Fact → Flag Dimension, with the flag dimension on the 1-side and the fact on the many-side. This allows the corresponding flag attributes to filter and group the relevant fact without creating direct relationships between unrelated business processes.

10.5 Date Relationships

The date dimension also acts as a conformed dimension across the applicable processes.

Fact/Snapshot	Date Column	dim_date Relationship	Status
fact_billing	bill_date	* : 1	Active
fact_patient_diagnostics	test_date	* : 1	Active
fact_drug_inventory	last_restock_date	* : 1	Active
snapshot_admission	admission_date	* : 1	Active
snapshot_admission	discharge_date	* : 1	Inactive

The inactive discharge-date relationship is used when analysis specifically requires filtering admissions by discharge date.

10.6 Relationship Design Principles

The relationship structure follows these principles:

1. No direct fact-to-fact relationships.
2. Dimensions generally filter facts through one-to-many relationships.
3. Flag/junk dimensions are connected only to their relevant facts.
4. dim_admission provides shared admission context across multiple processes.
5. Dimension-to-dimension relationships are used deliberately where required by the model.

6. Inactive relationships are used where multiple date roles or specialized relationships would otherwise create ambiguity.
7. Filter propagation is not forced simply to make every KPI respond to every slicer.

The model was subsequently tested for cardinality, filter direction, active/inactive status, ambiguity, grain and cross-fact behavior.

10.7 Key Modeling Characteristics

1. Grain-first design

Each fact was designed around a clearly defined business grain. This prevents measures from being unintentionally duplicated when different processes have different levels of detail.

2. Conformed dimensions

Common dimensions allow different business processes to be analyzed using consistent attributes. For example, patient and admission context can be used across billing, diagnostics and prescriptions.

3. No fact-to-fact relationships

Facts remain independent and are analyzed through shared dimensions or dim_admission, rather than being directly connected to one another.

4. Selective snowflaking

The model is not forced into a pure star schema. Relationships such as dim_admission → dim_beds → ward/department context, and the separate doctor/employee structure, are retained where they provide a better fit for the source data and analytical requirements.

5. Admission as shared analytical context

dim_admission represents the hospital encounter and provides a common filtering context for multiple processes. It is not a fact table and not a parent fact.

6. Accumulating snapshot

snapshot_admission tracks the admission lifecycle using available milestone dates and admission-level summaries such as diagnostic and prescription counts. Child-level data was aggregated to admission grain before being incorporated.

7. Date dimension

dim_date provides the common calendar and fiscal-time analysis required by the operational facts and snapshot.

10.8 Model Validation Status

The relationship structure has been tested in Power BI. The completed tests confirmed the intended cardinalities, filter directions, active/inactive relationships and absence of direct fact-to-fact relationships. The model therefore provides the structural foundation for the subsequent usage and business-scenario testing (Section 11).

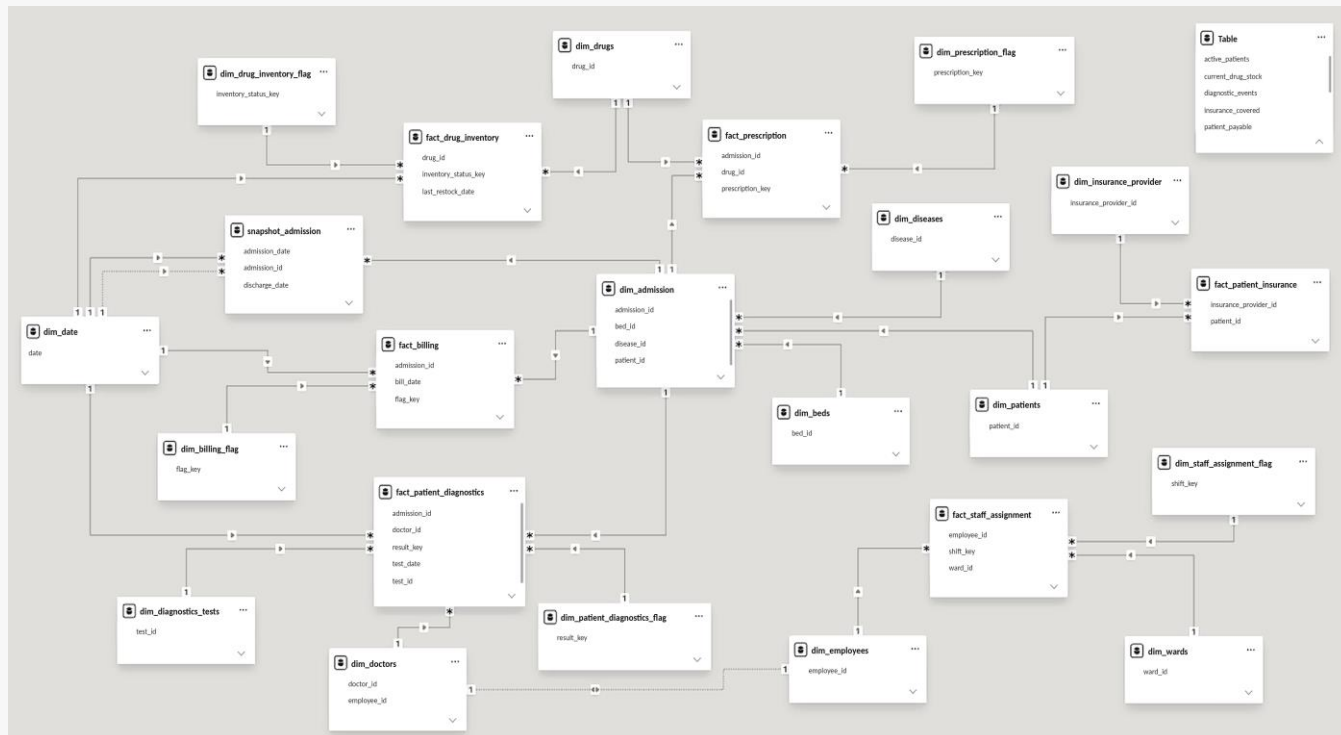


Figure 4. Final HMIS dimensional data model in Power BI.

11. Model Testing & Validation

After completing the dimensional model, the model was validated at three levels:

1. Data-level validation — comparing Power BI results against calculations made directly from the source CSV files.
2. Relationship validation — checking cardinality, filter direction, active/inactive relationships and filter propagation.
3. Business-logic validation — confirming that KPIs behave according to their business process and grain.

The objective was not only to verify that the model produces correct totals, but also to ensure that filters do not create unintended paths or double counting.

11.1 Ground-Truth KPI Validation

A set of baseline KPIs was calculated directly from the source CSV files using Excel and used as the reference for Power BI.

KPI	Source	Ground Truth
Total Patients	patient	30,000
Total Admissions	admission	45,000
Total Bills	billing	45,000
Total Billing Amount	billing	₹1,684,246,109
Insurance Covered Amount	billing	₹976,880,867
Patient Payable Amount	billing	₹707,365,242
Diagnostic Events	patient_diagnostic	63,269
Prescriptions	prescription	73,109
Insurance Policy Records	patient_insurance	21,617
Distinct Insured Patients	patient_insurance	18,000
Current Drug Stock	drug_inventory	130,878
Staff Assignments	staff_assignment	207
Employees	employee	500
Doctors	doctor	98

The corresponding Power BI cards matched these baseline values.

✓ **Result: PASS** — core measures showed no obvious duplication or aggregation errors at the overall model level.

11.2 Billing Measure Reconciliation

The three main billing measures were independently checked: total billed amount, insurance-covered amount, and patient-payable amount. The source data satisfies:

Insurance Covered + Patient Payable = Total Billing Amount → ₹976,880,867 + ₹707,365,242 = ₹1,684,246,109

The same relationship was confirmed in the Power BI model.

✓ **Result: PASS — the three financial measures can be used together without requiring billing_detail.**

11.3 Admission-Type Filter Validation

The dim_admission[admission_type] filter was tested using both major admission types.

Emergency

KPI	Expected	Power BI
Admissions	18,077	18,077
Billing Amount	₹673,597,695	₹673,597,695
Insurance Covered	₹387,554,733.70	₹387,554,733.70
Patient Payable	₹286,042,961.30	₹286,042,961.30
Diagnostic Events	25,378	25,378
Prescriptions	29,332	29,332

Elective

KPI	Expected	Power BI
Admissions	26,923	26,923
Billing Amount	₹1,010,648,414	₹1,010,648,414
Insurance Covered	₹589,326,133.30	₹589,326,133.30
Patient Payable	₹421,322,280.70	₹421,322,280.70
Diagnostic Events	37,891	37,891
Prescriptions	43,777	43,777

✓ **Result: PASS — validated the intended filter path from dim_admission to Billing, Diagnostics and Prescriptions.**

11.4 Department, Ward and Bed Validation

The hospital-location hierarchy was also tested. The model uses different grains deliberately:

Department → Ward → Bed → Admission

A department-level test using ICU produced:

KPI	Expected	Power BI
Admissions	4,040	4,040
Bills	4,040	4,040

KPI	Expected	Power BI
Billing Amount	₹149,425,049	₹149,425,049
Insurance Covered	₹85,840,114.40	₹85,840,114.40
Patient Payable	₹63,584,934.60	₹63,584,934.60
Diagnostic Events	5,545	5,545
Prescriptions	6,509	6,509

Ward-level and bed-level filtering was subsequently tested through the Power BI model.

✓ **Result: PASS** — the deliberately retained dimension-side hierarchy functions without creating unwanted fact duplication.

11.5 Patient Filter Validation

A patient-level filter was tested to verify that the shared patient context could reach the relevant business processes. For example, filtering to a single patient produced 3 admissions, 3 bills, 3 diagnostic events and 3 prescriptions — the corresponding facts responded consistently to the patient selection.

✓ **Result: PASS**

Modeling observation: a dimension does not automatically need to be filtered in reverse by every downstream fact. An admission-level filter does not necessarily need to reduce the overall patient dimension count — this depends on the intended analytical direction of the model.

11.6 Clinical Dimension Validation

Clinical dimensions were tested against their corresponding facts:

- **Disease** — selections correctly affected admission-related and downstream clinical measures.
- **Doctor** — selections correctly filtered diagnostic activity.
- **Diagnostic Test** — selections correctly filtered diagnostic events.
- **Drug** — selections correctly affected prescription and inventory analysis.

These tests confirmed that the clinical dimensions were connected to the appropriate business-process facts without requiring direct fact-to-fact relationships.

✓ **Result: PASS**

11.7 Cross-Fact Isolation Testing

The model was tested to ensure that filtering one business process did not unintentionally propagate into unrelated processes. For example, filtering a drug should affect fact_prescription and fact_drug_inventory, but should not unexpectedly filter admissions or billing. Similarly, admission-type filtering correctly affected admission-related facts but did not incorrectly change unrelated inventory totals.

✓ **Result: PASS** — confirms the value of avoiding direct fact-to-fact relationships.

11.8 Date Dimension Validation

The conformed dim_date was tested against the facts to which date relationships were intentionally established. The testing confirmed that date filtering operates according to the configured date relationships.

A date filter does not automatically make every KPI a time-based KPI. The user must select the appropriate fact and date role for the business question.

- Billing by year → use the billing fact and bill date.
- Diagnostic activity by year → use diagnostic events and test date.
- Admission analysis → use the admission/snapshot date relationship appropriate to the question.
- Prescription analysis by date cannot currently be performed using a prescription date because the source does not contain one.

✓ **Result: PASS**

11.9 Accumulating Snapshot Validation

snapshot_admission was validated at the admission grain. The snapshot contains one row per admission and incorporates admission-level summaries derived from child processes. Child-level data was aggregated before being merged into the snapshot to avoid fan-out:

patient_diagnostic → GROUP BY admission_id → test_count → snapshot_admission

prescription → GROUP BY admission_id → prescription_count → snapshot_admission

This preserves the required grain: 1 row = 1 admission.

✓ **Result: PASS**

11.10 Relationship Configuration Validation

The Power BI relationship configuration was reviewed for cardinality, cross-filter direction, active/inactive status, dimension-to-fact relationships, deliberate dimension-to-dimension relationships, potential ambiguous paths, and fact-to-fact relationships. The following principles were confirmed:

- Fact tables generally receive filters from dimensions.
- Single-direction filtering is used wherever practical.
- The dim_admission → snapshot_admission relationship was implemented with single-direction filtering rather than bidirectional 1:1 filtering.
- The deliberately inactive Doctor → Employee relationship remains inactive.
- Multiple date roles are handled using active/inactive relationships rather than activating competing paths simultaneously.
- No direct fact-to-fact relationships are used.

✓ **Result: PASS**

11.11 Measure and Grain Validation

A technically correct model can still produce an incorrect business answer if the wrong table or measure is selected.

For example, the question "How many admissions are there?" must first be interpreted in terms of its business process and required grain. Similarly, "How many bills were generated?" should use the billing fact, while "How many diagnostic events occurred?" should use the diagnostic fact. Therefore, KPI selection must follow:

Business Question → Business Process → Required Grain → Fact / Dimension → Measure → Filter / Date Role

This rule becomes especially important when users build their own Power BI visuals.

11.12 Validation Summary

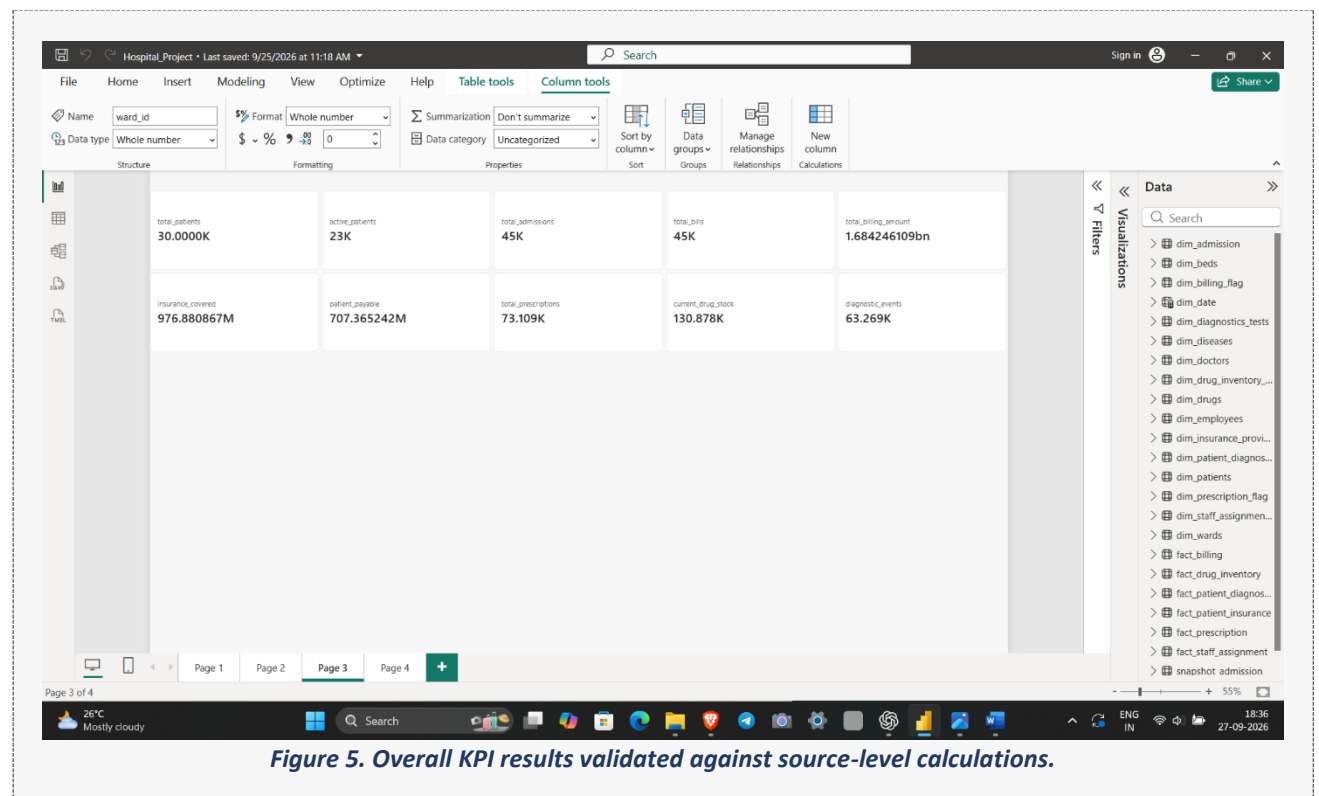
Validation Area	Result
Source-to-Power BI KPI reconciliation	PASS
Billing reconciliation	PASS
Admission-type filtering	PASS
Department filtering	PASS
Ward/Bed filtering	PASS
Patient filtering	PASS
Disease filtering	PASS
Doctor filtering	PASS
Diagnostic-test filtering	PASS
Drug filtering	PASS
Cross-fact isolation	PASS
Date filtering	PASS
Accumulating snapshot	PASS
Cardinality	PASS
Filter direction	PASS
Active/inactive relationships	PASS
Fact-to-fact relationship check	PASS
Grain/double-counting checks	PASS

Validation was performed by independently calculating expected results from the original source CSV files and comparing them with the corresponding Power BI measures and visual outputs. The validation covered overall KPIs, categorical filters, department/ward filtering, patient-level filtering, billing reconciliation, grain and double-counting, relationship behavior, cross-fact isolation, and admission snapshot calculations.

The independent validation calculations were performed during the modeling process using the original source data; the resulting validation findings are documented in this report.

Overall conclusion: the dimensional model passed the completed structural, numerical and business-behavior validation checks. It provides a reliable foundation for analytical reporting, subject to the documented source-data limitations and correct selection of facts, measures and date roles.

Representative figures for this section:



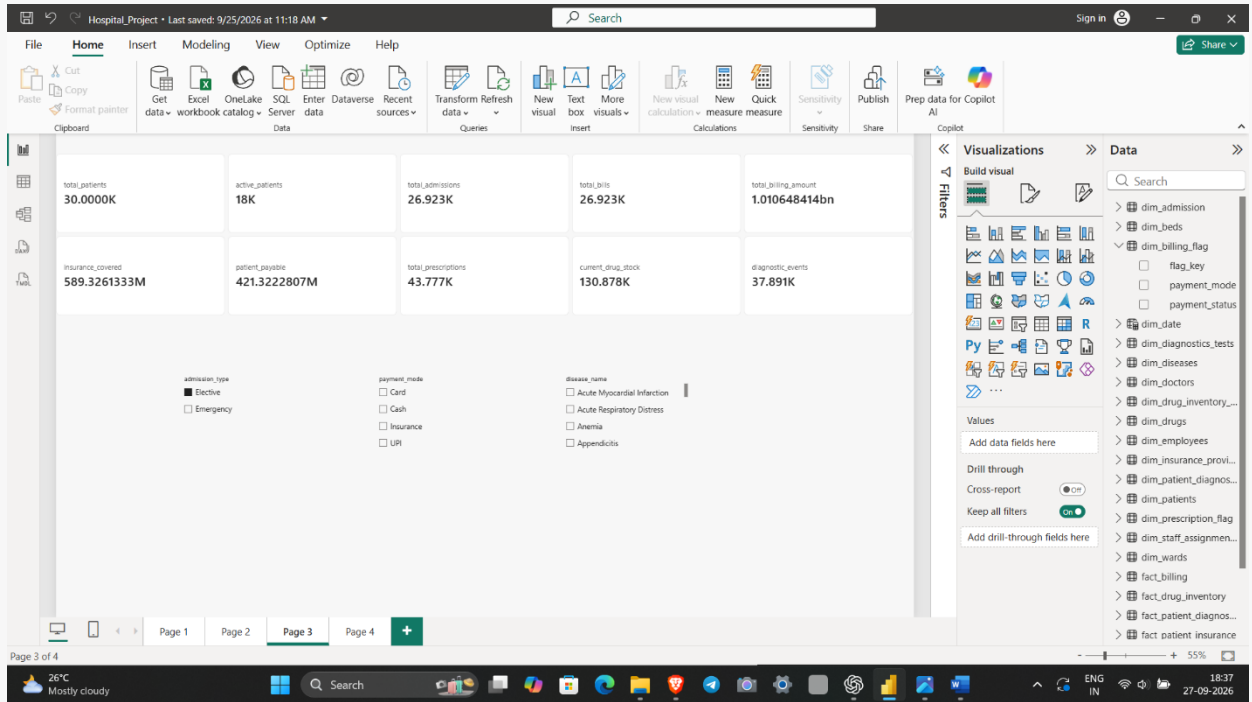


Figure 6. Validation of admission-type filter propagation across business-process facts.

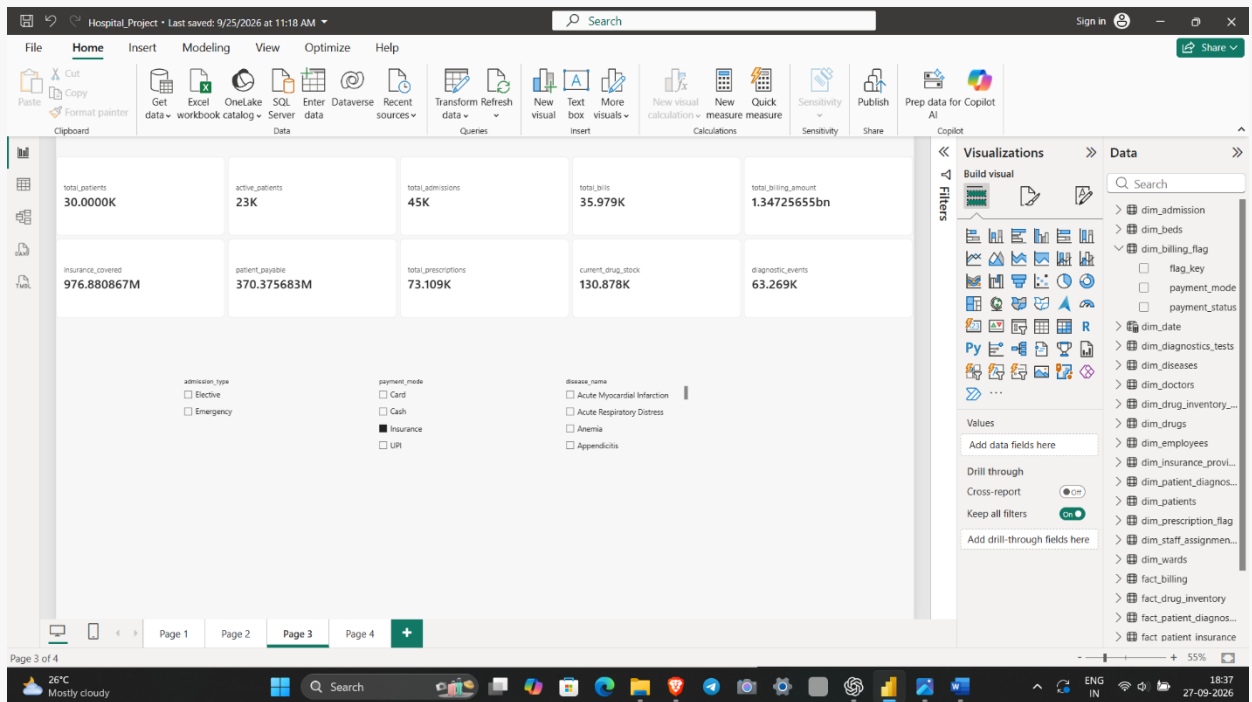


Figure 7. Validation of payment mode filter propagation across business-process facts.

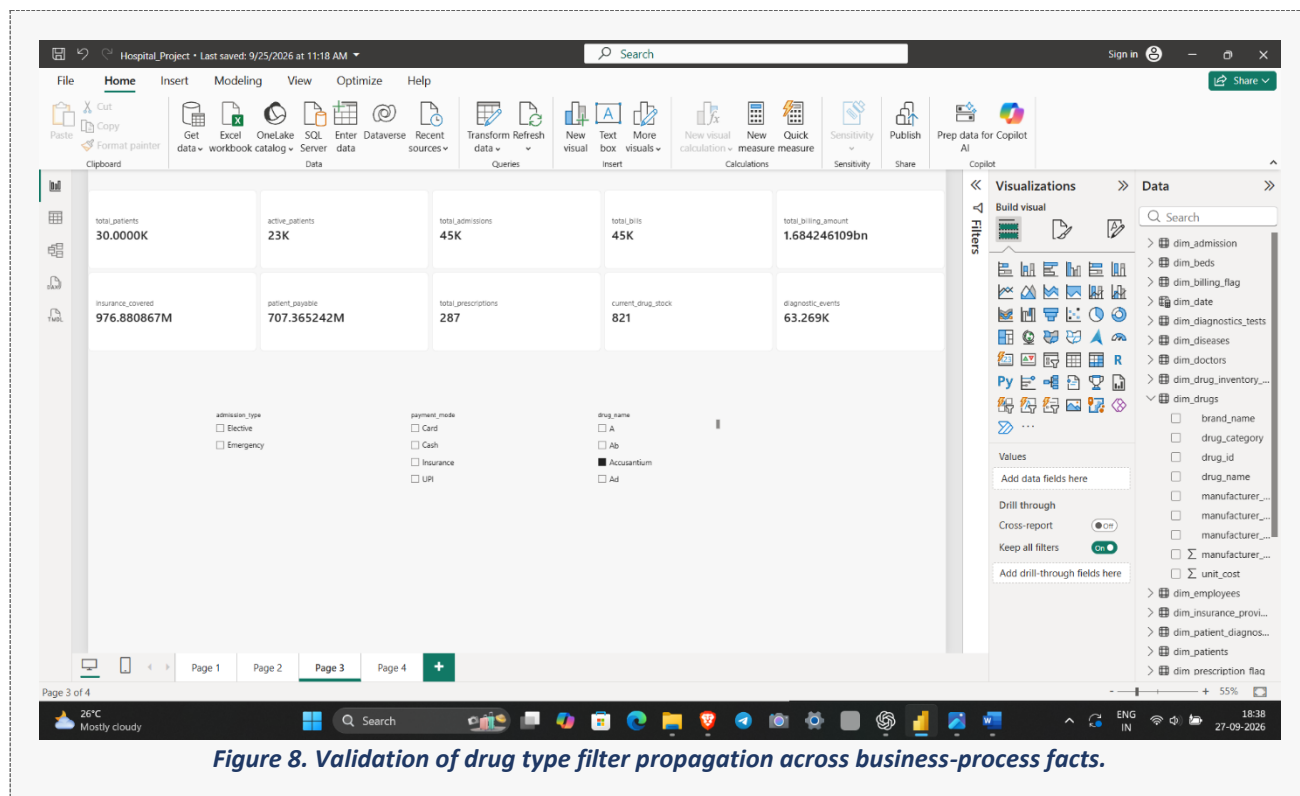


Figure 8. Validation of drug type filter propagation across business-process facts.

12. Data Model Usage Guide

Part 1 — Understanding the Model & Selecting the Right Data

Focus: before building anything, how should an analyst understand this model and decide where the answer should come from?

12.1 Purpose and Intended Users

The HMIS dimensional model is designed as a hospital business intelligence model that can support analysis across multiple hospital functions rather than serving a single department. The model therefore acts as a common analytical layer for hospital business intelligence.

Stakeholder	Typical Use
Hospital Management	Overall hospital performance and cross-functional analysis
Operations	Admissions, departments, wards, beds and operational activity
Finance	Billing, payments and insurance-covered amounts
Clinical / Medical Teams	Diseases, diagnostics, doctors and clinical activity
Insurance Team	Insurance providers, policies and coverage
Pharmacy / Inventory	Drugs, stock levels and reorder conditions
HR / Staff Management	Employees, doctors, ward assignments and shifts
BI / Data Analysts	Dashboard development, reporting and deeper analysis

12.2 Read the Model Before Building Reports

Key usage principle: the analyst must understand the data model before creating calculations, visuals or dashboards.

Before using the model, the analyst should study:

- The purpose of each table
- The grain of each table
- Fact and dimension classification
- Cardinality of relationships
- Filter direction
- Active and inactive relationships
- Which dimensions connect to which facts
- The purpose of dim_admission
- The role of snapshot_admission
- Available standardized DAX measures

This matters because the model contains several business processes with different grains. Billing is represented at bill grain, diagnostics at diagnostic-event grain, prescriptions at prescription grain, while

snapshot_admission represents the admission journey at one row per admission. Selecting a table simply because it contains a related field or number can therefore produce an incorrect business result.

12.3 Business Question → Data Selection Framework

Before creating a visual or writing DAX, follow this eight-step sequence:

- **Step 1 — Identify the business question.** Clearly define what needs to be answered. Example: "How much did the hospital bill in FY2025?"
- **Step 2 — Identify the business process.** Determine which hospital process the question belongs to (e.g. Billing).
- **Step 3 — Identify the required grain.** Determine what one analytical record represents (e.g. one bill).
- **Step 4 — Select the appropriate fact/dimension.** e.g. fact_billing
- **Step 5 — Use the standard measure if available in the designed Model.** e.g. Total Billing Amount
- **Step 6 — Identify the correct date.** e.g. bill_date
- **Step 7 — Apply relevant dimensions/filters.** e.g. Fiscal Year from dim_date
- **Step 8 — Validate the result.** Confirm that the calculation represents the original business question.

The overall workflow:

Business Question → Business Process → Required Grain → Fact / Dimension → Standard Measure → Date / Date Role → Relevant Filters → Validate → Build Report

This approach helps prevent a common BI problem: a calculation can be technically correct but still answer the wrong business question.

12.4 KPI and Fact Selection Guide

The model contains separate facts for different hospital processes. The analyst should select the fact according to the business question.

Business Requirement	Primary Table	Typical Information
Billing amount	fact_billing	Total billing, insurance covered, patient payable
Bill count	fact_billing	Number of bills
Admissions	snapshot_admission	Admission-level journey
Admission attributes	dim_admission	Admission type, status and admission context
Diagnostic activity	fact_patient_diagnostics	Diagnostic events, results, doctors and tests
Prescription activity	fact_prescription	Drugs, dosage, frequency and duration
Current inventory	fact_drug_inventory	Current stock, reorder level and inventory status
Insurance policies	fact_patient_insurance	Policy and coverage information
Staff assignments	fact_staff_assignment	Employee-to-ward assignment and shift

Business Requirement	Primary Table	Typical Information
Admission journey / LOS	snapshot_admission	Admission and discharge-related journey information

These are guidelines based on the intended business use of the model. The analyst should still examine the model and business question before selecting a table.

12.5 Admission-Specific Usage Rules

Admission analysis requires particular attention because both dim_admission and snapshot_admission are available for different analytical purposes.

Requirement	Recommended Source	Approach
Total number of admissions	snapshot_admission	Count admission records
Active patient/admission analysis	dim_admission	Use admission dimension
Admissions by admission type	dim_admission	Group/filter by admission type
Admissions by disease	dim_disease → dim_admission	Filter admission context through disease
Admissions by department	dim_beds → dim_admission	Filter admission context through bed/department
Admissions by year	dim_date → snapshot_admission	Use active admission-date relationship
Length of stay	snapshot_admission	Use admission journey information
<i>"Admissions" is not a single universal calculation for every business question — the correct source depends on what the analyst is trying to measure.</i>		

For example, the model's total-admission KPI and an admission-dimension analysis may use different tables because they answer slightly different analytical questions.

12.6 Standardized KPI Measures

A set of standard DAX measures has been created and stored separately in an empty measure table. Examples include:

- Total Billing Amount
- Total Bills
- Total Admissions
- Diagnostic Events
- Insurance Covered
- Total Patients
- Total Prescriptions
- Other standard business measures

The purpose of these measures is to establish consistent KPI definitions across different users and departments. For example, if two departments require total billing, both should use the same standardized Total Billing Amount measure rather than independently creating their own versions.

Usage principle: use an existing standardized measure whenever the required KPI is already available. Create new DAX only when the business requirement genuinely requires a new calculation.

Part 1 is therefore about one central idea: before touching Power BI visuals or DAX, understand the model and determine exactly where the answer should come from.

Part 2 — Applying the Model Correctly

Focus: now that the analyst knows where the data comes from, how should relationships, dates, DAX and filters be used correctly?

12.7 Filter and Relationship Usage

The model should be used according to its defined relationship structure. Before creating a visual or calculation, the analyst should inspect the dimension-to-fact relationships, cardinality, filter direction, and active/inactive status.

Use the existing relationship path first. Use DAX only when the business requirement or relationship configuration requires it.

For example:

- dim_drugs → fact_prescription / fact_drug_inventory
- dim_doctors → fact_patient_diagnostics
- dim_diseases → dim_admission
- dim_beds → dim_admission
- dim_patients → admission-related analysis
- dim_insurance_provider → fact_patient_insurance

Not every dimension is intended to filter every fact. For example, filtering a drug should affect prescription or inventory analysis, but it should not be expected to change unrelated hospital admission or billing measures.

A KPI not changing under a particular slicer is not automatically a model error. The expected behaviour depends on the relationship path and the business question.

12.8 Date and DAX Usage

dim_date is the common date dimension for the processes where an appropriate source date exists. The current date relationships include:

Table	Date Column	Relationship
fact_billing	bill_date	Active
fact_patient_diagnostics	test_date	Active
snapshot_admission	admission_date	Active
snapshot_admission	discharge_date	Inactive
fact_drug_inventory	last_restock_date	Active

Calendar Year vs Fiscal Year

The model provides both a Calendar Year (January to December) and a Fiscal Year used for business/accounting analysis. The analyst should select the appropriate one based on the business requirement.

- "Billing by financial year" → use Fiscal Year.
- "Admissions by calendar year" → use Calendar Year with the appropriate admission date.

Inactive Date Relationship

discharge_date is an inactive relationship to dim_date. Therefore, a normal date slicer does not automatically use discharge date for analysis. If the business question is "How many admissions were discharged in each year?", the analyst must explicitly use the discharge-date relationship through appropriate DAX rather than assuming the active admission-date relationship answers the question.

12.9 Business Examples / Case Studies

Case 1 — Finance: Billing by Fiscal Year

Question: *"What was the hospital's total billing amount by fiscal year?"*

Use:

- **Fact** → fact_billing
- **Measure** → Total Billing Amount
- **Date** → dim_date
- **Date field** → bill_date
- **Time attribute** → Fiscal Year

The analyst should not calculate official billing totals from billing_detail.

Case 2 — Operations: Admissions by Department

Question: *"How many admissions were associated with each department?"*

Use:

- **Admission context** → dim_admission
- **Department** → dim_beds
- **Relationship path** → dim_beds → dim_admission
- **Admission count** → appropriate admission calculation

The relationship path should be checked before building the visual.

Case 3 — Clinical Management: Diagnostic Activity

Question: *"Which departments or doctors are handling more diagnostic activity?"*

Use:

- **Fact** → fact_patient_diagnostics
- **Measure** → Diagnostic Events
- **Doctor** → dim_doctors
- **Diagnostic test** → dim_diagnostic_tests
- **Relevant department context** → according to the model's relationships
- **Date** → test_date

This type of analysis can also support resource-management decisions, such as identifying areas with higher diagnostic workload.

Case 4 — Pharmacy: Low Inventory

Question: “Which drugs currently require attention?”

Use:

- **Fact** → fact_drug_inventory
- **Drug** → dim_drugs
- **Attributes** → current_stock, reorder_level, inventory_status
- **Filter** → Low inventory

This represents the current inventory state. It should not be interpreted as historical stock movement because the source does not contain inventory movement history.

Case 5 — Management: Patient Length of Stay

Question: “How long are patients staying in the hospital?”

Use:

- **Table** → snapshot_admission
- **Grain** → One row per admission
- **Measure/attribute** → Length of Stay
- **Relevant dimensions** → Patient, disease, bed/department, admission attributes
- **Date** → Admission Date for admission-based analysis

If the question instead concerns discharge timing, the analyst must use the inactive discharge-date relationship appropriately.

12.10 Important Usage Cautions

- **1. Do not create fact-to-fact relationships.** The model was deliberately designed to avoid direct fact-to-fact relationships. Facts should be analysed through their appropriate dimensions and relationship paths.
- **2. Do not expect every slicer to affect every KPI.** A dimension should only affect a fact when an appropriate relationship path exists — for example, Drug → Prescriptions/Inventory does not mean Drug → Admissions/Billing. This is intentional model behaviour.
- **3. Use fact_billing for official billing measures.** The official financial measures should come from fact_billing. billing_detail is retained outside the final semantic model because its detail-level amounts did not directly reconcile with the bill-level totals. It can potentially be incorporated later if charge-level analysis becomes a requirement and its business rules are clearly established.
- **4. Do not invent missing dates.** fact_prescription has no prescription date in the source, so prescription analysis cannot legitimately be presented as a prescription-date trend. Similarly, fact_staff_assignment has no assignment date, so historical staff-assignment trends cannot be inferred from this model.

- **5. Understand the insurance limitation.** fact_patient_insurance can be used for insurance policies, insured patients, insurance providers, coverage percentage and policy dates. However, there is no reliable source relationship linking an individual bill/admission to a specific insurance policy — analysts should not assume a billing transaction belongs to a particular policy.
- **6. Understand the inventory limitation.** fact_drug_inventory represents current inventory information: current stock, reorder level, inventory status and last restock date. It does not provide historical stock movement.
- **7. Maintain the flag/junk dimensions.** Some flag/junk dimensions were created using surrogate keys representing combinations of attributes. As new source data enters the warehouse, new combinations may appear; if these are not added to the corresponding flag/junk tables, filters may not work correctly, records may not map correctly, and reports may produce incomplete or incorrect numbers.

This is an important maintenance responsibility when new data is loaded into the warehouse.

12.11 Quick Decision Framework and Analyst Checklist

The following decision framework can be used whenever a new analysis is required:

Business Question → Business Process → Required Grain → Fact / Dimension → Standard Measure → Required Date → Relevant Filters → Check Relationship Path → Use DAX if Needed → Validate → Build the Visual

Before Publishing a KPI or Dashboard

- ☐ I understand the overall data model.
- ☐ I understand the grain of the relevant table.
- ☐ I identified the correct business process.
- ☐ I selected the appropriate fact/dimension.
- ☐ I used the standard KPI where available.
- ☐ I selected the correct date and date role.
- ☐ I checked cardinality and filter direction where relevant.
- ☐ I confirmed that the required filter path exists.
- ☐ I have not created an unintended fact-to-fact analysis.
- ☐ I have not assumed every slicer should affect every KPI.
- ☐ I have not invented information that does not exist in the source.
- ☐ I validated the result against the intended business definition.
- ☐ If new source data was loaded, I checked the flag/junk dimensions for new combinations.

Final usage principle: read and understand the data model first. Then follow the usage rules in this manual to select the appropriate fact, dimension, measure, date and filter path. Finally, validate the result against the business question before using it in a dashboard or decision.

13. Limitations and Future Improvements

Although the HMIS dimensional model successfully supports the defined hospital business intelligence requirements and passed the model validation tests, several limitations originate from the structure and availability of the source data. These limitations should be understood before extending the model or interpreting its outputs.

13.1 Current Limitations

13.1.1 No Historical Bed Occupancy Data

The source contains bed information such as bed status, ward and department, but does not provide historical bed occupancy events or occupancy-period records. Therefore, the model can support analysis of beds, wards, departments and current bed-related information. However, it cannot reliably calculate historical metrics such as bed occupancy rate, bed utilization over time, daily occupied beds, average bed occupancy, or historical bed turnover. A future version would require a dated bed-occupancy or bed-status history.

13.1.2 Inventory Is Current-State Data

fact_drug_inventory represents the current inventory state, including current stock, reorder level, inventory status and last restock date. The source does not contain historical stock movement transactions. Therefore, the model cannot reliably answer questions such as daily/monthly stock levels, historical stock consumption, stock movement over time, opening vs closing inventory, or inventory turnover. A future inventory model would require a dedicated inventory movement or periodic snapshot fact.

13.1.3 No Historical Staff Assignment

fact_staff_assignment records employee-to-ward assignments and shifts, but the source does not contain an assignment date or effective-date history. Therefore, the model can analyse available staff assignments and shifts but cannot reliably provide historical staffing levels, staffing by date, staff movement between wards over time, or historical workload by shift. Adding effective dates or a dated staff scheduling/assignment process would enable this analysis.

13.1.4 No Prescription Date

The source prescription table does not contain a prescription date. Consequently, fact_prescription cannot be connected to dim_date for prescription-date analysis. The model can still analyse prescription counts, drugs prescribed, dosage, frequency, duration, and prescriptions by admission or other relevant dimensions. However, analysts should not interpret these as prescription trends over time.

13.1.5 Limited Insurance Policy-to-Billing Linkage

The insurance data contains patient policies, providers, coverage percentages and policy dates. However, the source does not provide a reliable relationship connecting a particular billing transaction or admission to a specific insurance policy. Therefore, the model supports insurance policy and provider analysis, but should not be used to make unsupported claims such as “this particular bill was covered under this specific policy.” A future implementation would benefit from a reliable insurance claim or billing-policy transaction linking the policy to the relevant admission/bill.

13.1.6 Billing Detail Is Not Included in the Semantic Model

billing_detail contains charge-level information, but its amounts did not directly reconcile with billing.total_amount in the source data. Because the current business requirements did not require charge-level analysis, and because the relationship between the detail amounts and official bill totals was not sufficiently established, the table was retained for traceability but excluded from the final semantic model. This means the current model does not provide detailed charge-type reporting through the semantic layer. If charge-level analysis becomes a future requirement, the business rules and reconciliation logic should first be established before incorporating the table.

13.1.7 Source Date Range Observation

Although the dataset was described as covering 2020–2025, profiling showed that some operational dates extend into January 2026 — for example, discharge, billing and diagnostic dates extend beyond December 2025. The date dimension was therefore designed using the actual operational date range required by the model rather than relying only on the stated dataset description. This should be considered when interpreting time-based reports.

13.1.8 Synthetic Dataset

The HMIS dataset is synthetic but designed to represent a realistic hospital environment. Therefore, the model demonstrates dimensional modeling, business intelligence and analytical design principles, but its business rules and data distributions should not automatically be treated as representative of a real hospital's operational system. A production implementation would require validation with actual hospital stakeholders, source-system documentation and real business definitions.

13.1.9 Flag/Junk Dimension Maintenance

Some flag/junk dimensions use surrogate keys to represent combinations of attributes. When new data is loaded, new combinations may appear that were not present when these tables were originally created. If these combinations are not added, filters may not work correctly or may produce incomplete results. Therefore, these dimensions require maintenance whenever the source data structure or value combinations change.

13.2 Future Improvements

The following enhancements could extend the analytical capabilities of the model.

Current Limitation	Potential Future Improvement
No historical bed occupancy	Create a dated bed occupancy/periodic snapshot fact
Current inventory only	Add inventory movement transactions or periodic inventory snapshots
No staff assignment history	Add effective dates and historical staff assignments
No prescription date	Add prescription/order date to the source
Limited insurance linkage	Add insurance claim/policy-to-bill linkage
Billing detail not used	Establish charge-level business rules and reconciliation before adding it
Limited historical tracking	Introduce appropriate SCD strategies where historical changes become relevant
Single-source environment	Integrate additional hospital source systems if required
Flag/junk maintenance	Automate detection and loading of new attribute combinations

13.3 Potential Future Analytical Capabilities

With the additional data described above, the model could eventually support more advanced hospital BI use cases, including:

- Historical bed occupancy and utilization
- Bed turnover and capacity planning
- Historical drug consumption and inventory forecasting
- Staff workload and staffing requirements over time
- Prescription trends
- Detailed insurance claims analysis
- Charge-level revenue analysis
- Historical tracking of changing hospital entities and attributes
- More comprehensive resource planning

These improvements should be driven by business requirements and source-data availability, rather than by adding dimensional modeling techniques without a corresponding analytical need.

14. Final Summary

This project developed a Hospital Management Information System (HMIS) dimensional data model in Power BI using practical Kimball dimensional modeling principles. The objective was to transform a normalized relational hospital dataset into an analytical model capable of supporting business questions across hospital operations, clinical activity, finance, insurance, pharmacy and staff management.

14.1 Modeling Approach

The model was developed using a grain-first and business-process-first approach. The major principles applied were:

- Identify the business process before designing the fact table.
- Define the grain of every fact and dimension.
- Separate facts from descriptive dimensions.
- Use conformed dimensions where multiple business processes require common analytical context.
- Avoid direct fact-to-fact relationships.
- Aggregate lower-grain data before combining it with higher-grain data.
- Flatten dimensions where this improves analytical usability.
- Retain selective dimension-side relationships where flattening would create unnecessary duplication, sparsity or table expansion.
- Use a star schema as the default, with selective snowflake structures where justified.

The resulting model is therefore a hybrid dimensional model, rather than a completely pure star schema.

14.2 Final Model

The final model contains dimensions, transactional facts and snapshot facts supporting the identified hospital processes. Components include:

Dimensions

- dim_patients
- dim_admission
- dim_beds
- dim_wards
- dim_diseases
- dim_doctors
- dim_employees
- dim_diagnostic_tests
- dim_drugs
- dim_insurance_provider
- dim_date
- dim_patient_diagnostics_flag

- dim_billing_flag
- dim_drug_inventory_flag
- dim_prescription_flag
- dim_staff_assignment_flag

Facts / Snapshots

- fact_billing
- fact_patient_diagnostics
- fact_prescription
- fact_patient_insurance
- fact_drug_inventory
- fact_staff_assignment
- snapshot_admission

The snapshot_admission table was designed as an accumulating snapshot, representing the admission lifecycle at one row per admission. Child-level information such as diagnostic events and prescriptions was aggregated to admission grain before being incorporated into the admission snapshot. This avoided fan-out and double-counting.

14.3 Major Modeling Decisions

Shared Admission Context

dim_admission was created as a genuine encounter-level dimension rather than simply using it as a workaround for fact-to-fact relationships. It provides admission-specific descriptive context that can be used across multiple hospital processes.

Selective Snowflake

The model deliberately retains some dimension-to-dimension relationships, such as the admission-related structure, instead of flattening everything into one large dimension. This was chosen where complete flattening would introduce unnecessary repetition, sparsity or table expansion.

No Fact-to-Fact Relationships

Facts independently connect to their appropriate dimensions or shared admission context. This prevents ambiguous analytical paths and reduces the risk of incorrect aggregation.

Billing Detail Exclusion

billing_detail was retained for traceability but excluded from the final semantic model because its measures did not directly reconcile with the official billing totals and charge-level analysis was not required by the current business requirements.

Date Dimension

A conformed dim_date was created and connected only to processes for which a meaningful date existed in the source. This avoided creating artificial date relationships for tables such as prescriptions, where no prescription date exists.

Standard DAX Measures

Standard KPI measures were created in a dedicated measure table so that commonly used metrics have consistent definitions across reports and dashboards.

14.4 Data Quality and Validation

The model was validated against the original CSV data rather than relying only on visual inspection. Core KPI validation included:

KPI	Validated Value
Patients	30,000
Admissions	45,000
Total Billing	₹1,684,246,109
Insurance Covered	₹976,880,867
Patient Payable	₹707,365,242
Diagnostic Events	63,269
Prescriptions	73,109
Current Drug Stock	130,878

The billing figures also reconciled:

Insurance Covered + Patient Payable = Total Billing → ₹976,880,867 + ₹707,365,242 = ₹1,684,246,109

Additional testing covered:

- Admission-type filtering
- Department filtering
- Ward/bed filtering
- Patient filtering
- Disease filtering
- Doctor filtering
- Diagnostic-test filtering
- Drug filtering
- Date relationships
- Cross-fact filter isolation
- Relationship cardinality
- Cross-filter direction
- Active/inactive relationships
- Grain and double-counting
- Admission snapshot validation

The validation confirmed that the model behaved consistently with the defined business requirements and source data.

14.5 Key Learning from the Project

Building a data model is not simply a matter of converting every source table into a fact or dimension.

The source system and the analytical model serve different purposes. A good analytical model requires understanding:

**Business Question → Business Process → Grain → Fact/Dimension → Measure → Date →
Relationships → Validation**

The project also demonstrated that a technically valid relationship does not automatically mean that every slicer should affect every KPI. The analyst must understand how the model is intended to be used before selecting tables, measures and filters.

14.6 Limitations

The model is constrained by the available source data. Major limitations include:

- No historical bed occupancy data.
- Inventory is current-state rather than historical movement data.
- Staff assignments do not contain assignment dates.
- Prescriptions have no prescription date.
- Insurance policies cannot be reliably linked to individual bills/admissions.
- Billing-detail amounts do not directly reconcile with billing totals.
- The dataset is synthetic.
- Some operational dates extend into January 2026 despite the stated 2020–2025 coverage.

These limitations were documented rather than being filled with assumptions during modeling.

14.7 Future Direction

The model can be extended when additional business requirements and source data become available. Potential extensions include:

- Bed occupancy history
- Inventory movement tracking
- Historical staff scheduling
- Prescription dates
- Insurance claim/policy-to-billing linkage
- Charge-level billing analysis
- Historical dimension tracking through appropriate SCD techniques
- Additional hospital source-system integration

These improvements should be introduced based on actual business requirements and available source data, rather than adding complexity unnecessarily.

14.8 Final Conclusion

This project demonstrates the practical application of Kimball dimensional modeling in Power BI, from source profiling and business requirements through dimensional design, transformation, relationship configuration and validation.

The final HMIS model provides a structured analytical foundation for hospital reporting while maintaining clear table grains, conformed analytical context and controlled filter propagation.

Understand the business → identify the grain → design the model → transform the data → validate the numbers → document how the model should be used.

The resulting model is not simply a collection of tables. It is a business-oriented analytical model designed to produce reliable and interpretable hospital insights from the available data.